

# Distributional Drift and Capability Retention in Small-Scale LLM Post-Training

Aayaann Kausar

Indian Institute of Technology Kharagpur, India



code and measurement artifacts

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## Abstract

Post-training changes a language model’s behaviour everywhere, not only where the task is evaluated. We report a controlled, reproducible study of how far the updated policy moves from its initialization—and what that costs—across three objectives: supervised fine-tuning (SFT) on matched in-distribution and expert-written targets, on-policy reinforcement learning with verifiable rewards (GRPO), and a small preference-optimization demonstration. All instruction-following arms share one base model (SmolLM2-135M-Instruct), one prompt distribution (a filtered 397-prompt IFEval subset), one held-out split and one set of rule-based verifiers; they differ only in where the supervision comes from and in the update rule. Drift is exact full-vocabulary forward KL,  $D_{KL}(\pi_0 \parallel \pi_\theta)$ , per token on the new task’s inputs; retention is held-out perplexity on two corpora plus a common-sense multiple-choice battery. At the checkpoints we obtained, GRPO reaches the highest held-out satisfaction ( $0.483 \pm 0.023$ , three seeds) while moving  $2.5\times$  less than either SFT arm (0.134 vs. 0.331/0.334 nats/token), and the SFT arms raise held-out perplexity by 10–21% where GRPO changes it by  $+0.3\%/+0.4\%$ . The same degraded checkpoints are *indistinguishable* from base on the accuracy battery ( $\leq 1$  point), so the choice of retention probe determines whether degradation is detected at all. A separate Countdown experiment (Qwen2.5-3B-Instruct) raises the sampled solve rate from 9.1% to 21.9% over 500 steps and solves 32/95 held-out puzzles greedily with every corresponding base answer wrong, yet 64% of the model’s intermediate candidate equations mis-evaluate; we therefore read the change as amplification of a behaviour the initialization already sampled, not as a new capability.

## 1 Introduction

Post-training turns a pretrained language model into an assistant, and it never edits the model locally. Supervised fine-tuning (SFT) maximises the likelihood of target responses everywhere the model is queried; preference optimization raises the relative likelihood of chosen over rejected responses; on-policy reinforcement learning from verifiable rewards (RLVR, our GRPO instantiation) instead resamples responses from the current policy and reweights them by reward. All three alter the model’s next-token distribution over *all* contexts, not merely the evaluation prompts that define the new task. The classical concern that post-training damages previously acquired behaviour is therefore well posed for language models, and the practical question is comparative: which update rule

pays less for the same gain?

**What is known and what is missing.** In-language fine-tuning can degrade safety behaviour [4] and produce broad behavioural change that task metrics do not capture. Shenfeld et al. [6] argue that forgetting is predicted by the KL divergence between the fine-tuned and base policies measured on the new task’s input distribution, and that on-policy RL is biased toward KL-minimal solutions. Chu et al. [1] contrast SFT, which memorises, with RL, which generalises across rule and visual variations; Yue et al. [7] show RLVR gains over the base model shrink as  $k$  grows in  $\text{pass}@k$ , i.e. improvement reflects more efficient sampling of already-reachable answers; Shao et al. [5] find that identical RLVR training can behave qualitatively differently across model families. For data choices, Pan et al. [3] show chosen-response quality dominates DPO and online DPO approaches SFT on chosen responses.

Three gaps motivate this paper. First, comparisons of update rules are usually reported through a single measurement instrument, while drift and retention are properties of the full distribution; a study that reports only the instrument supporting its headline is hard to interpret mechanically. Second, “RL preserves capability” is often reported at a scale out of reach for independent replication, so the mechanism is hard to probe. Third, many arms compared as “SFT vs. RL” differ in prompt count, epochs, learning rate and even data source, so the update rule is confounded with the data. We attack all three at a scale that fits one consumer GPU: a single initialization, a shared prompt distribution and split, matched-data supervised arms, several retention instruments, and an exact full-vocabulary drift measurement.

**Problem setting and notation.** Let  $\pi_0$  be the initialization (the base model) and  $\pi_\theta$  the post-trained policy;  $\pi_{\text{ref}}$  denotes the frozen reference used by preference and on-policy objectives, and  $\mathcal{V}$  the vocabulary. Prompts  $x$  are drawn from a task distribution  $\mathcal{D}$ , responses  $y = (y_1, \dots, y_T)$  from the policy, and  $c_t = (x, y_{<t})$  is the context at position  $t$ . We study three families of arms at matched data:

- **SFT-IN:** SFT on targets sampled from  $\pi_0$  itself and filtered by a verifier (best-of-K).
- **SFT-OUT:** SFT on expert-written targets for the same prompts.
- **RL:** GRPO on the same prompts with a rule-based reward and no target responses at all.

**Drift.** The quantity we measure is the forward (base-to-

trained) per-token KL on the task’s inputs, with the expectation taken over contexts the base policy actually visits:

$$D_{\text{KL}}(\pi_0 \parallel \pi_\theta) = \mathbb{E}_{x \sim \mathcal{D}} \mathbb{E}_{y \sim \pi_0(\cdot | x)} \left[ \frac{1}{T} \sum_{t=1}^T \text{kl}_t \right], \quad (1)$$

$$\text{kl}_t = D_{\text{KL}}(\pi_0(\cdot | c_t) \parallel \pi_\theta(\cdot | c_t)),$$

We use the forward direction because the completions are sampled from  $\pi_0$ ; the reverse direction would require samples from  $\pi_\theta$  to be estimated on the same contexts. The direction matters, the two are not equal, and (1) is the quantity used in every drift number reported below.

**Retention.** Retention is measured by two instruments of deliberately different sensitivity: held-out general-text perplexity,  $\text{PPL}(\pi_\theta; \text{corpus})$ , and a top-1 multiple-choice accuracy basket (HellaSwag, ARC, PIQA). Neither is a capability measurement, which is itself part of the finding.

**Research questions.** **RQ1** At matched data, does the update rule change task skill and drift jointly? **RQ2** Does the *source* of SFT targets—in-distribution versus expert—drive the drift? **RQ3** Do retention instruments agree about the same checkpoints? **RQ4** Does verifiable-reward RL improve reasoning by acquiring new behaviour or by amplifying behaviour the initialization already sampled?

**Contributions.** (i) A controlled arm design in which supervised and on-policy arms share initialization, prompt pool, held-out split and verifier family, with supervised prompts matched to the on-policy prompt set by index (Algorithm 1). (ii) An exact per-token forward KL estimator with the direction fixed and justified. (iii) A dual-instrument retention analysis in which the instruments disagree, reported as a result rather than as noise. (iv) A negative result: the in-distribution/off-distribution distinction does not move the drift needle here. (v) A conservative reading of a Countdown reasoning gain as amplification, supported by the base model’s own chains and by the arithmetic error rate in the trained model’s intermediate steps. Every number in this paper traces to a committed artifact (Appendix A); no new training was run in its preparation.

**Scope.** Two small models, two task families, one model family per scale, three seeds on some axes and one on others, and checkpoints that are not matched on task performance. The paper therefore supports scoped statements about these runs, and states explicitly in Section 6 which broader claims it does not support. Two follow-ups would sharpen it: a drift-matched supervised control, and a large-k sampled base-model evaluation on the same puzzles.

## 2 Method

### 2.1 Objectives

**Supervised fine-tuning** maximises the likelihood of target tokens under a completion-only mask, i.e. only positions inside the target response contribute:

$$\mathcal{L}_{\text{SFT}}(\theta) = -\mathbb{E}_{(x, y^*) \sim \mathcal{D}_{\text{SFT}}} \left[ \frac{1}{|y^*|} \sum_{t=1}^{|y^*|} \log \pi_\theta(y_t^* | c_t^*) \right]. \quad (2)$$

The two supervised arms differ only in  $\mathcal{D}_{\text{SFT}}$ : SFT-IN uses verifier-filtered samples from  $\pi_0$  (best-of-K), SFT-OUT uses expert-written responses. Both maximise likelihood on strings drawn from a distribution that is not  $\pi_\theta$ , which is the structural difference from on-policy learning: the objective assigns mass to specific continuations rather than reweighting continuations the policy already produces.

**Preference optimization** maximises the log-ratio margin over pairs, with the reference policy as an anchor:

$$\mathcal{L}_{\text{DPO}}(\theta) = -\mathbb{E}_{(x, y_w, y_l)} \log \sigma(\beta[\Delta_w - \Delta_l]), \quad (3)$$

$$\Delta_y = \log \frac{\pi_\theta(y|x)}{\pi_{\text{ref}}(y|x)}.$$

Differentiating gives a gradient proportional to  $\sigma(-\beta[\Delta_w - \Delta_l]) \beta(g_w - g_l)$ , where  $g_y = \nabla_\theta \log \pi_\theta(y | x)$ : it constrains the *difference* of the two log-likelihoods against the reference and places no term that keeps  $\log \pi_\theta(y_w | x)$  from decreasing, so both chosen and rejected likelihoods can fall together. This is the mechanism behind the collapse reported in §6 and it is a property of the objective, not a claim about the run.

**On-policy RL with verifiable rewards** (GRPO) samples a group of  $G$  responses per prompt from the frozen step policy  $\pi_{\theta_{\text{old}}}$ , assigns each a scalar reward  $r_i$ , and maximises a clipped, group-normalised objective with a KL penalty to  $\pi_{\text{ref}}$ :

$$\mathcal{J}_{\text{GRPO}}(\theta) = \mathbb{E} \left[ \frac{1}{G} \sum_{i=1}^G \frac{1}{T_i} \sum_{t=1}^{T_i} \min(\rho_{i,t} \hat{A}_i, \text{clip}(\rho_{i,t}, 1 \pm \epsilon) \hat{A}_i) - \beta_{\text{KL}} D_{\text{KL}}(\pi_{\theta_{\text{old}}} \parallel \pi_{\text{ref}}) \right], \quad (4)$$

with  $\rho_{i,t} = \pi_\theta(y_{i,t} | c_{i,t}) / \pi_{\theta_{\text{old}}}(y_{i,t} | c_{i,t})$  and  $\hat{A}_i = (r_i - \text{mean}(r)) / \text{std}(r)$ . The objective is identically zero for a prompt whose group has equal rewards, so gradients exist only where sampling already separates better from worse responses; and the  $1/T_i$  normalisation dilutes the penalty on long responses when  $\hat{A}_i < 0$ , which is the length bias analysed by Liu et al. [2]. The  $T_i$  remains in the mask, not in a learned direction.

**Rewards.** The instruction-following reward is the mean of per-constraint indicators, gated to zero below a distinct-3-gram fraction threshold  $\tau = 0.35$  and with explicit anti-echo rules:

$$r_{\text{ife}}(y) = \mathbb{1}_{\left[ \frac{\#\{3\text{-grams}\}}{|y|} \geq \tau \right]} \cdot \frac{1}{|\mathcal{C}|} \sum_{c \in \mathcal{C}} \mathbb{1}_{[c \text{ satisfied}]}, \quad (5)$$

The Countdown reward scores only the final expression, with a graded proximity term:

$$r_{\text{cd}}(y) = w_{\text{corr}} \mathbb{1}_{[\text{value}(y) = \text{target}]} + w_{\text{fmt}} \mathbb{1}_{[\text{format}]} + w_{\text{prox}} \max\left(0, 1 - \frac{|\text{value}(y) - \text{target}|}{\text{target}}\right), \quad (6)$$

with  $(w_{\text{corr}}, w_{\text{prox}}, w_{\text{fmt}}) = (2.0, 0.5, 0.2)$ . Rewards are part of the treatment, not controls: an earlier farmable design was replaced rather than ablated, and no reward ablation exists.

### 2.2 Measuring drift exactly

The estimator behind (1) is exact and cheap at this scale: for a sampled completion we evaluate both policies over the full vocabulary at every position, which avoids the plug-in bias

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**Algorithm 1** Matched in-distribution / off-distribution target construction

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**Require:** initialization  $\pi_0$ ; filtered IFEval pool  $\mathcal{D}_{\text{pool}}$ ; count  $n$ ; best-of- $K$ ; satisfaction threshold  $\gamma$

**Ensure:** paired datasets  $\mathcal{D}_{\text{in}}, \mathcal{D}_{\text{out}}$  over an identical index set  $S$

- 1:  $S \leftarrow \{i : \pi_0 \text{ has } \geq 1 \text{ completion passing all constraints}\}$
- 2: **for** all  $i \in S$  **do**
- 3:    $\{y_i^{(k)}\}_{k=1}^K \leftarrow \text{sample } K \text{ completions from } \pi_0(\cdot | x_i) \text{ at } T=1.0$
- 4:    $c_{\text{in}} \leftarrow \arg\max_k \text{verifier}(x_i, y_i^{(k)})$   $\triangleright$  targets that pass
- 5:    $c_{\text{out}} \leftarrow \text{expert answer for } x_i$
- 6:    $S \leftarrow S \setminus \{i\}$  if either target fails the verifier  $\triangleright$  identical prompts
- 7: **end for**
- 8:  $S \leftarrow S_{1:n}; \quad \mathcal{D}_{\text{in}} \leftarrow \{(x_i, c_{\text{in}})\}, \mathcal{D}_{\text{out}} \leftarrow \{(x_i, c_{\text{out}})\}$
- 9: run both arms for the same epochs, batch and optimizer budget
- 10: **return**  $\mathcal{D}_{\text{in}}, \mathcal{D}_{\text{out}}$

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**Algorithm 2** Exact forward-KL drift per token on held-out task prompts

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**Require:** initialization  $\pi_0$ ; post-trained  $\pi_\theta$ ; held-out prompt set  $\mathcal{D}_{\text{held}}$ ;  $N_s$  completions per prompt

**Ensure:**  $\widehat{D}_{\text{KL}}$  (nats per token), coverage, mean completion length

- 1:  $\text{acc} \leftarrow 0; \quad n_{\text{tok}} \leftarrow 0; \quad \ell_{\text{sum}} \leftarrow 0$
- 2: **for** all  $x \in \mathcal{D}_{\text{held}}$  **do**
- 3:   **for**  $s = 1$  **to**  $N_s$  **do**
- 4:      $y \leftarrow \text{sample from } \pi_0(\cdot | x)$   $\triangleright$  contexts = base-policy trajectories
- 5:     **for** positions  $t$  with a non-padding token in the completion **do**
- 6:       compute  $\pi_0(v | c_t)$  and  $\pi_\theta(v | c_t)$  exactly over  $\mathcal{V}$  with  $T=1.0$
- 7:        $\text{acc} += \sum_v \pi_0(v | c_t) [\log \pi_0(v | c_t) - \log \pi_\theta(v | c_t)]$
- 8:        $n_{\text{tok}} += 1$
- 9:     **end for**
- 10:    $\ell_{\text{sum}} += |y|$
- 11:   **end for**
- 12: **end for**
- 13: **return**  $\widehat{D}_{\text{KL}} = \text{acc}/n_{\text{tok}}, \quad \bar{T} = \ell_{\text{sum}}/\text{completions}$

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of single-token estimates, and we average the per-position divergences over positions and completions:

$$\text{kl}_t = \sum_{v \in \mathcal{V}} \pi_0(v | c_t) [\log \pi_0(v | c_t) - \log \pi_\theta(v | c_t)],$$

$$\widehat{D}_{\text{KL}} = \frac{1}{N_s} \sum_{s=1}^{N_s} \frac{1}{T_s} \sum_{t=1}^{T_s} \text{kl}_t. \quad (7)$$

Two properties matter for interpretation. The reported quantity is a *per-token* average, so arms with different completion-length distributions are not compared on a reparameterisation of the same sequence-level quantity; and the tokens are realisations, so  $\widehat{D}_{\text{KL}}$  is unbiased for the distributional quantity defined by the sampled contexts.

**Evaluation protocol.** Task skill, drift and retention are computed on disjoint instruments over a single frozen split: prompts are partitioned once with a fixed seed, every method is evaluated on that partition, so no difference between arms can come from re-selecting the evaluation set. Skill is greedy satisfaction on held-out prompts; retention is PPL on two held-out corpora plus the accuracy basket; drift is Algorithm 2 with  $N_s = 4$  completions per held-out prompt, sampled from the base policy.

### 3 Experimental Setup

**Models and task.** All instruction-following arms start from SmolLM2-135M-Instruct (an instruction-tuned model, so satisfaction of an instruction-following benchmark is a measurable property rather than a floor) and train on google/IFEval.

**Table 1:** Instruction-following arms (base SmolLM2-135M-Instruct; prompt pool 317/80; 11 verified instruction types). Effective batch is per-device batch  $\times$  gradient accumulation. Hyperparameters are read from the committed training-guide text; only the seeds differ between otherwise identical rows.

|                       | in-dist<br>SFT | off-dist<br>SFT | GRPO<br>(RL) |
|-----------------------|----------------|-----------------|--------------|
| Source                | own b-of-16    | expert          | rule reward  |
| Prompts               | $\approx 165$  | $\approx 165$   | 317          |
| Targets satisfy       | yes            | yes             | —            |
| Learning rate         | 5e-5           | 5e-5            | 1e-5         |
| Epochs                | 15             | 15              | 3            |
| Batch $\times$ accum. | $8 \times 4$   | $8 \times 4$    | 32           |
| Group size $G$        | —              | —               | 16           |
| KL coeff. $\beta$     | —              | —               | 0.005        |
| Sampling temp.        | —              | —               | 1.0          |
| Completion cap        | —              | —               | 200          |
| Precision             | bf16           | bf16            | bf16         |
| Training seeds        | 3              | 3               | 3            |

**Table 2:** Countdown arms (base Qwen2.5-3B-Instruct); evaluated once per epoch over 100 puzzles. Wall-clock is estimated from the first logged step and the total step count.

|                      | GRPO                              |
|----------------------|-----------------------------------|
| Objective            | r=16, $\alpha=32$ , q/k/v/o + MLP |
| LoRA                 | (2.0, 0.5, 0.2)                   |
| Reward weights       | (2.0, 0.5, 0.2)                   |
| Learning rate        | 2e-5                              |
| Epochs / steps       | 1 / 500                           |
| Batch / accum.       | $2 \times 16$                     |
| Effective batch      | 64                                |
| Completion cap       | 1024                              |
| Sampling temperature | 1.0                               |
| Rollout engine       | vLLM (colocated)                  |
| Train / eval puzzles | 2000 / 100                        |
| Time per step        | $\approx 44$ s                    |
| Estimated wall clock | $\approx 6$ h                     |

The repository’s verifier implements 20 of the 25 upstream instruction types and drops examples using any unimplemented type. Re-executing that filter on the public dataset reproduces 397 retained prompts, of which 161 carry more than one constraint; a fixed split gives 317 training and 80 held-out prompts (seed 42), shared by every arm. The reasoning experiment uses Qwen2.5-3B-Instruct and Jiayi-Pan/Countdown-Tasks-3to4.

**Matched prompts and supervision.** The supervised arms train on the  $\approx 165$  prompts for which both an expert target and a base-model best-of-16 sample that fully satisfies the verifier exist: the intersection of the two prompt sets is the training set for both supervised arms, so the arms differ only in the provenance of the target text. GRPO uses all 317 training prompts and needs no targets. The number of matched prompts is not stored in a committed artifact—the implementation prints it at run time—so it is treated as the least well-pinned number in this paper (Appendix A). Table 1 gives the configurations and Table 2 the Countdown configuration.

**Provenance of the numbers.** The repository excludes committed per-run result files from version control (trainer\_output/\*\_runs.json are git-ignored), so the values reported here are recovered from committed figures and logs in three ways: means are read from the printed annotations of the committed figures; standard deviations are recovered by axis-calibrated digitisation of the same figures (linear-fit residuals

below  $5 \times 10^{-4}$  in data units, cross-checked against the printed means); and everything inside the Countdown study is computed directly from the two committed logs. Appendix A lists every quantity, its route, and what a re-run must re-measure.

## 4 Results

### 4.1 RQ1: task skill and drift are not in tension here

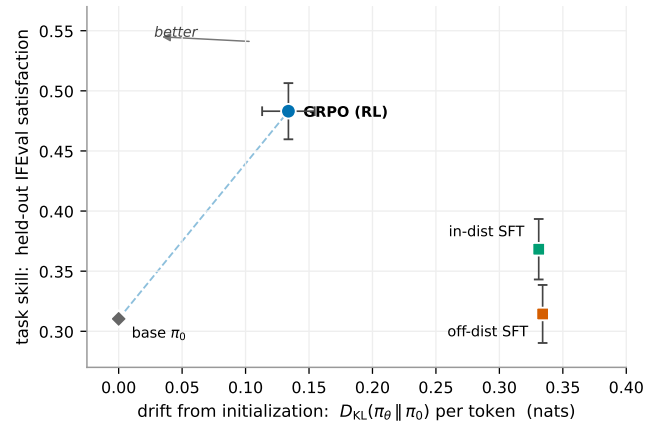
Table 3 and Figure 1 give the primary comparison. The observation is that in this configuration the on-policy arm is not trading one axis against the other: at the end of its budget it attains the highest held-out satisfaction ( $0.483 \pm 0.023$  over three seeds, versus  $0.368 \pm 0.025$  in-distribution and  $0.315 \pm 0.024$  off-distribution SFT) while exhibiting the smallest measured drift from the initialization ( $0.134 \pm 0.021$  versus 0.331 and 0.334 nats per token). Both comparisons lie outside the respective seed spreads, which carry no measurable spread in the committed figure for the supervised drift values.

Three qualifications are part of this result, not caveats appended to it. First, the arms are compared at the ends of their training budgets, not at matched task performance: in-distribution SFT is evaluated at a lower held-out task score than GRPO, so the data do not separate “the on-policy rule moves less per unit of task skill acquired” from “the supervised runs were still short of the task when they stopped.” That is the reason we do not title this finding “RL forgets less”. Second, the supervised and on-policy arms do not share the number of optimization steps ( $\approx 77$  versus 474), the learning rate ( $5 \times 10^{-5}$  versus  $1 \times 10^{-5}$ ), or the form of the supervision signal, so the comparison should be read as one update rule versus another *under a shared task and evaluation*, and not as an isolation of the update rule from the supervision signal. Third, the drift axis is a scalar summary of a per-position divergence field, normalised by response length: the ordering is therefore under the evaluated model, data, budgets and measurement protocol, and we do not claim a general law.

### 4.2 RQ2: the source of supervised targets does not drive the drift

The two supervised arms are the controlled experiment. They share the base model, the matched prompt set ( $\approx 165$  prompts), the example count, the epochs (15), the learning rate, the effective batch and therefore the optimizer budget ( $\approx 77$  steps). They differ only in whether the target completion was sampled from the base model and filtered by the verifier (in-distribution) or written by an external expert (off-distribution). Table 3 shows that this manipulation changed drift by 1% of its own magnitude (0.331 versus 0.334 nats per token, i.e. 0.003 apart), while the GRPO arm differs from both by 0.20 nats; the retention probes agree that the two arms are equivalent (+10.6% versus +10.4% on wikitext-2, +20.7% versus +19.4% on pile-10k, with seed spreads at or below 0.2 percentage points), and the multiple-choice difference is in the opposite direction (0.495 versus 0.502).

Two features of the panel are worth recording separately. With the *larger* drift and the *lower* task skill, off-distribution SFT has the *highest* multiple-choice accuracy (0.502): a probe that tracks drift or capability loss would not order the arms



**Figure 1:** Task skill against distributional drift at the end of training, base SmolLM2-135M-Instruct. Horizontal axis is the exact forward  $D_{KL}(\pi_0 \parallel \pi_\theta)$  per token on held-out IFEval prompts sampled from the shared base policy; vertical axis is held-out IFEval satisfaction under greedy decoding. Error bars are  $\pm 1$  standard deviation over three training seeds. The GRPO point has a measurable horizontal error bar ( $\pm 0.021$ ); the supervised points show no measurable horizontal spread in the committed figure, and their plotting bounds do not reach the x-axis digitisation tolerance, so they should be read as single measurements rather than as seed means. The dashed line joins the base model to the guide point and is not a measured trajectory; no intermediate checkpoint of the GRPO run was probed for drift.

that way. And the ceiling is low by construction: the base model scores 0.493, so the instrument’s resolution is bounded by how much room a 135M model has left, and the repository notes that it deliberately excluded GSM8K because the base model is already at a floor of about 1.4%: the model’s weakest and most relevant skill is unmeasured.

The result is a negative one for the hypothesis that supervised drift is driven principally by the off-distributionness of the targets. It is not a refutation of that hypothesis in general, and it does not identify the mechanism: the two supervised arms are also matched on the number of passes over a small target set, which is a plausible common cause of memorisation-driven drift, and the in-distribution arm’s targets are a verifier-filtered high-reward tail rather than an unbiased sample of the base model’s own output.

### 4.3 RQ3: the two retention probes disagree

The two instruments do not tell the same story, and the disagreement is informative. Held-out perplexity orders the arms exactly as drift does: on wikitext-2, base 10.6%, in-distribution SFT +10.4% relative, off-distribution SFT +10.4%, GRPO +0.3%; on pile-10k, +20.7%, +19.4% and +0.4% respectively, with per-seed spreads at or below 0.2 percentage points. Across the mixture of the two corpora the separation is unambiguous (Figure 2a). The commonsense multiple-choice battery separates nothing: 0.492 to 0.502 against a base of 0.493, a spread of under one point, with the *highest* score belonging to the arm with the largest drift and lowest task score (Figure 2b).

We read this as two functionals of the same checkpoints rather than as evidence that one probe is wrong. Perplexity sums over the full conditional distribution and is sensitive to any redistribution of probability mass, exactly the quantity

**Table 3:** Post-training at the end of each training budget (base SmoLM2-135M-Instruct). *Task skill* is held-out IFEval satisfaction (%) under greedy decoding over  $n = 80$  prompts. *Drift* is the exact forward  $D_{KL}(\pi_0 || \pi_\theta)$  per token on held-out prompts, sampled from the base policy.  $\Delta$ PPL is the relative increase in held-out perplexity on two corpora, and the multiple-choice score is the mean of HellaSwag, ARC and PIQA (*acc\_norm*). Skill, perplexity and drift are means over three training seeds; the drift entries for the supervised arms are single measurements, and the multiple-choice entries are single runs per arm, as indicated by  $n$ . Values are recovered from the committed figures (Appendix A); digitisation uncertainty is  $\pm 0.002$  on drift and  $\pm 0.01$  on skill.

| Arm          | Task skill $\uparrow$               | $n$ | Drift $\downarrow$                  | $n$ | $\Delta$ PPL wikitext | $\Delta$ PPL pile | MC acc |
|--------------|-------------------------------------|-----|-------------------------------------|-----|-----------------------|-------------------|--------|
| base $\pi_0$ | 0.310                               | 1   | 0                                   | —   | —                     | —                 | 0.493  |
| IN-DIST SFT  | $0.368 \pm 0.025$                   | 3   | 0.331                               | 1   | +10.6%                | +20.7%            | 0.495  |
| OFF-DIST SFT | $0.315 \pm 0.024$                   | 3   | 0.334                               | 1   | +10.4%                | +19.4%            | 0.502  |
| GRPO (RL)    | <b><math>0.483 \pm 0.023</math></b> | 3   | <b><math>0.134 \pm 0.021</math></b> | 3   | <b>+0.3%</b>          | <b>+0.4%</b>      | 0.492  |

**Table 4:** Countdown GRPO training log averaged over successive blocks of 20 logged points (step increment by 5, so each block spans 100 optimizer steps). All columns are derived from the committed per-step log. KL is the trainer’s own estimate and not the forward divergence of (7). Length dips early and recovers rather than inflating (see §6).

| Logged steps | Solve | Prox. | Format | Reward | KL     | Entropy | Length |
|--------------|-------|-------|--------|--------|--------|---------|--------|
| 5–100        | 0.119 | 0.242 | 0.884  | 0.535  | 0.0111 | 1.110   | 476.1  |
| 105–200      | 0.238 | 0.409 | 0.941  | 0.868  | 0.0315 | 0.987   | 473.9  |
| 205–300      | 0.249 | 0.430 | 0.941  | 0.901  | 0.0361 | 1.015   | 495.7  |
| 305–400      | 0.266 | 0.451 | 0.936  | 0.944  | 0.0354 | 1.043   | 510.0  |
| 405–500      | 0.237 | 0.425 | 0.936  | 0.873  | 0.0382 | 1.091   | 520.9  |

drift measures; the multiple-choice score depends only on the identity of the top-ranked option, on prompts from a different task family, and any change that preserves the ordering of the plausible options is invisible to it. A 20% perplexity increase with zero accuracy change is a coherent state of affairs: the model can keep its preferences between candidate answers while spreading mass differently across the tail. The measurement rule the data support is therefore that a retention claim needs at least two probes, because a null result from a coarse battery is not evidence of preserved capability—and conversely, retention loss that matters behaviourally has to be measured directly.

#### 4.4 RQ4: verifiable-reward RL amplifies a search behaviour it does not invent

The 3B Countdown experiment raises the solve rate of the policy’s own samples from 9.1% at the first logged step to 21.9% at step 500 (peak 37.8% around step 230), with the proximity component of (6) rising earlier and further and format adherence saturating near 0.96; the trainer’s own KL estimate against the reference grows from 0.0005 to 0.0344, and entropy falls from 1.25 to 1.09, consistent with a policy concentrating on a workable strategy rather than broadening. Zero-variance groups—prompts whose G samples tie—are almost absent (at most 5% of groups at any logged step), so the reward signal stayed alive throughout. Table 4 averages the series over successive blocks of 20 logged points.

On the held-out set, the committed comparison log enumerates 95 puzzles and reports the trained checkpoint solving 32 of them greedily, a rate of 33.7% (Wilson 95% interval 25.0–43.7%), against 18 of the corresponding base-model answers that contain no parseable expression and none that evaluate correctly. Because the logging script returns before printing whenever GRPO fails, the base model is only observable on those 32 puzzles and its rate over the full 95 is unmeasured,

so the panel in Figure 3(b) is one-sided by construction.

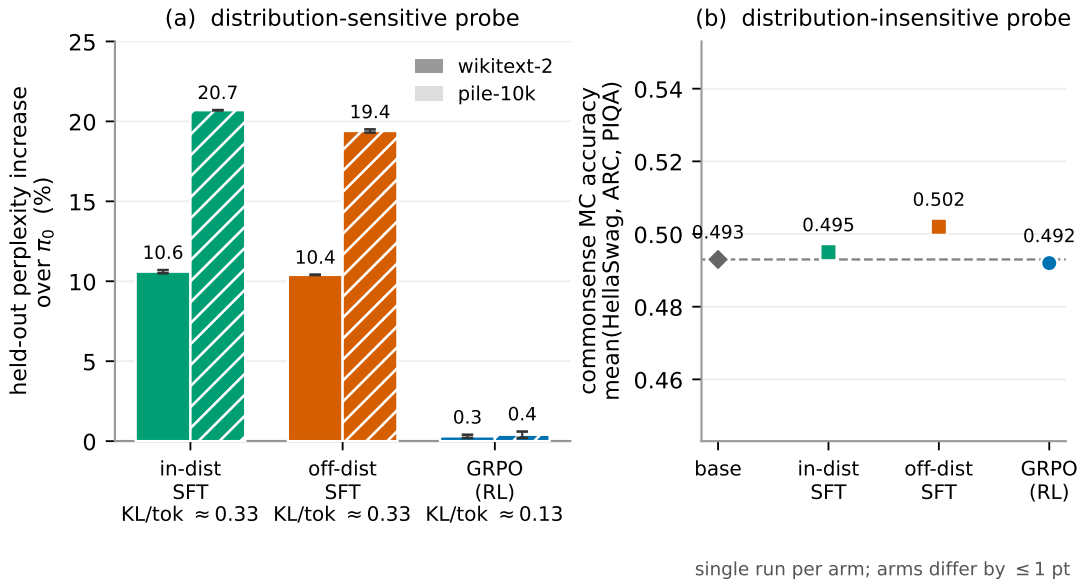
The behavioural evidence is the chains themselves, and it is what makes the RQ the interesting one. The base model already produces the qualitative form of the search—the phrase “let’s try another” occurs 99 times in its chain and 142 times in the trained chain—and the trained chain shows the same structure with better execution: it enumerates candidate expressions, labels them “too high” or “too low”, and reaches a correct final answer. Three observations make the conservative reading the right one. First, the observed improvement is concentrated in search and format rather than in arithmetic: of 28 parsed candidate equations written by the trained model in the prompt-level examples, 18 (64%) mis-evaluate their own expression. Second, the reward (6) scores only the final expression, so nothing in the objective provides a direct pressure toward correct intermediate arithmetic. Third—and decisively for the wording—the base model’s chain contains the search behaviour the training then stabilises, which matches the account in Shenfeld et al. [6] and Yue et al. [7]: we characterise the change as *amplification of an already-sampled search-and-verify behaviour*, not as acquisition of a new capability. This is a one-model, one-task, no-large-k-sampling observation: we can show a behaviour became stable and productive, and we cannot bound what the initialization could already do.

## 5 Discussion

We keep three levels of statement separate. *Observations* are what was measured; *inferences* are what the measurements support within the tested configuration; *hypotheses* are candidate mechanisms the experiment does not isolate.

### 5.1 What was measured

- (i) At the end of their training budgets the three instruction-following arms are ordered by both task skill and drift: the on-policy arm attains the highest held-out satisfaction and the smallest forward KL per token.
- (ii) The two supervised arms, differing only in target provenance at fixed prompts, epochs and budget, are indistinguishable in drift and retention.
- (iii) Held-out perplexity separates the arms by 10–21% on two corpora with seed spreads at or below 0.2 percentage points, while the commonsense battery places all arms within one point of the base model and orders the largest-drift arm highest.
- (iv) In the Countdown run the sampled solve rate rises  $2.4\times$  over training, the held-out greedy rate is 32/95, and most of the model’s intermediate candidate arithmetic is wrong.
- (v) An earlier, farmable Countdown reward scored about 1.5



**Figure 2:** Two retention probes applied to the same checkpoints. (a) Relative increase in held-out word perplexity over the base model on wikitext-2 and the first 500 documents of pile-10k. Bars are means over three training seeds and error bars are  $\pm 1$  standard deviation (in-distribution SFT:  $+10.6 \pm 0.1\%$  and  $+20.7 \pm 0.0\%$ ; off-distribution SFT:  $+10.4 \pm 0.0\%$  and  $+19.4 \pm 0.1\%$ ; GRPO:  $+0.3 \pm 0.1\%$  and  $+0.4 \pm 0.2\%$ ); per-arm drift is printed beneath each label. (b) Commonsense multiple-choice accuracy on the same checkpoints under  $\text{acc\_norm}$  (HellaSwag, ARC, PIQA; single run per arm, no seed spread). The distribution-sensitive probe separates the arms with a wide margin while all four models remain within one point of each other on the distribution-insensitive probe, and the arm with the largest drift and lowest task score has the *highest* multiple-choice score.

points for well-formed-but-wrong expressions; the reward we study is the repair, and it is part of the treatment rather than a control. Two implementation findings also belong to the evidence: preference optimization at the supervised learning rate ( $5 \times 10^{-5}$ ) collapses the 135M model into incoherent text within one epoch where  $5 \times 10^{-6}$  does not — consistent with (3), which constrains only the difference of log-ratios — and the committed log carries the loader’s own warning that the chat template’s end-of-turn token encodes to a different id than `eos_token_id`, a defect that affects base and trained evaluations symmetrically.

## 5.2 What the measurements support

**The retention claim, scoped.** Within the tested configuration the observed retention cost is a property of the update rule as much as of the task. The perplexity penalty paid by the supervised arms is not a general property of learning a new task: the on-policy arm learned a harder version of the same task—higher held-out satisfaction on the same prompts—while moving  $2.5\times$  less and changing general-text perplexity by less than half a percent. Nothing here licenses “on-policy RL forgets less than SFT”, because the arms are unmatched on skill, steps, learning rate and supervision signal.

**The instruments measure different functionals.** A 20% perplexity increase with zero accuracy change is coherent: perplexity aggregates the full conditional distribution and detects any redistribution of mass, whereas the accuracy battery depends only on the identity of the top-ranked option on a different task family. The practical rule is that a retention claim needs at least two probes of understood sensitivity, and that a null result from an accuracy battery is not evidence of

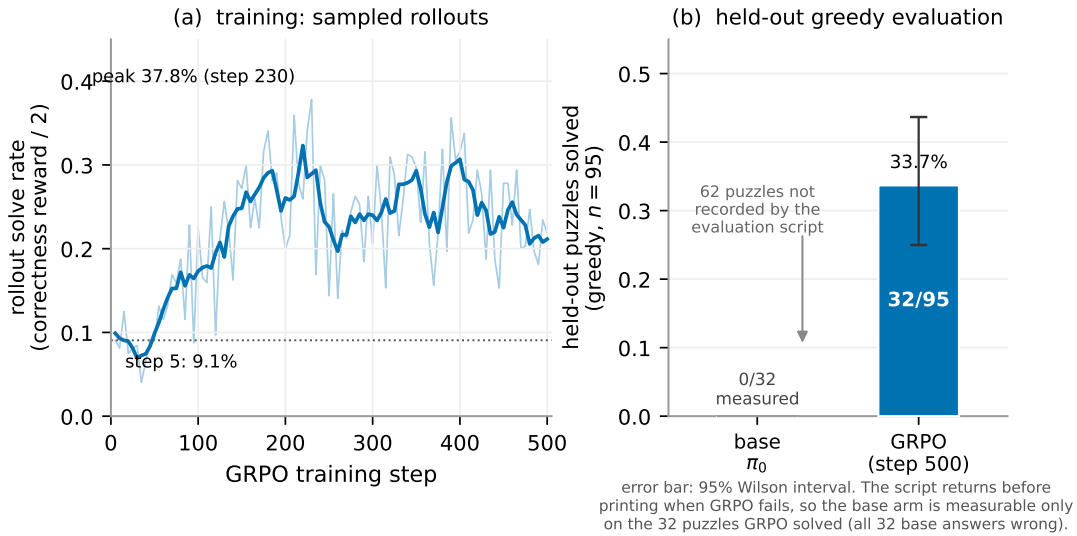
preserved capability.

**The data-source hypothesis fails here, and that is informative.** Drift from supervised fine-tuning in this configuration is not explained by whether targets lie inside or outside the base model’s distribution. Whatever produced 0.33 nats is common to both arms; the most natural candidate is the number of passes over a small target set, and a second candidate is that the in-distribution targets are a verifier-filtered high-reward tail rather than an unbiased sample of  $\pi_0$ .

**The length bias did not appear in the aggregate.** The  $1/T_i$  normalisation of (4) dilutes the penalty on long negative-advantage responses and thus predicts length inflation. The Countdown series is non-monotone: it dips to 416.8 tokens at step 90 and recovers to 520.9 (mean of the last 20 logged points) against 476.1 for the first 20, while the first and last individual points fall from 550.7 to 531.7. Net growth of about 9% over the run is not runaway inflation, but the logged average covers all rollouts, so it cannot isolate the negative-advantage subset in which the bias acts; we report this as a non-materialisation in the aggregate, not as a demonstration that the bias is absent.

## 5.3 Candidate mechanisms (hypotheses)

**H1 (already-sampled mass).** The GRPO gradient is zero for a prompt whose rollouts all receive the same reward, and the objective reweights the likelihood of sampled tokens rather than assigning probability to unseen strings, so a policy optimised this way can move only in directions its own samples open. This is the mechanism proposed by Shenfeld et al. [6] and it is consistent with both experiments, including the Countdown



**Figure 3:** Countdown, base Qwen2.5-3B-Instruct, 500 GRPO steps. (a) Solve rate of the policy’s own sampled rollouts during training, computed as the logged correctness reward divided by its weight 2.0 in (6) (faint line: raw per-logged-value series; solid line: centred 5-point rolling mean). The dotted line is the first logged value, 9.1%; at the step-500 checkpoint the solve rate is 21.9% and the peak is 37.8% at step 230. (b) Held-out greedy evaluation: of the 95 puzzles enumerated in the committed log, the trained policy solves 32 (33.7%, Wilson interval 25.0–43.7%). The bar labelled *base* is hatched because the log’s own scanning script returns before printing whenever GRPO fails, so the base model is evaluable on only the 32 puzzles GRPO solved, all of which it answers wrongly; the base solve rate over the full set is unmeasured, and the panel is not to be read as a 33.7%-versus-0% comparison.

result where the base model already emits the search structure. Our design does not isolate it: that needs matched task skill across arms plus a sweep of the degree of off-policy-ness with reward and data held fixed.

**H2 (drift, not objective).** Under this hypothesis the supervised arms retain worse because they moved further, and a supervised arm constrained to the same drift budget would retain as well. The natural experiment—a KL-regularised or early-stopped supervised arm tuned to reach 0.134 nats per token—is not available in the artifact. Our data are consistent with H2 and cannot distinguish it from the alternative that supervised objectives damage something the drift summary does not capture.

**H3 (amplification within existing coverage).** The evidence of §4 is consistent with an account in which verifiable-reward training raises the probability of a solution path the initialization can already sample, with the base model’s coverage as the ceiling—the claim Yue et al. [7] test with large-k sampling. We do not test it: with one model, one task and greedy evaluation we can show a behaviour became stable but cannot bound what was available before training.

#### 5.4 Implications for practitioners

For this class of small-scale post-training, the on-policy run with a rule-based reward was cheaper on both axes at once—although the arms are unmatched on supervision cost, since the reward-based run needed no target construction. Drift measured with an exact full-vocabulary divergence on task prompts was computable offline for a 135M model with four samples per held-out prompt and is a more direct signal of global change than task accuracy. And because the two retention probes disagreed, a practitioner should choose the probe whose sensitivity matches the damage they fear, and measure any specific behaviour they care about directly.

## 6 Limitations

Each limitation below corresponds to a specific property of these experiments.

**Scale and models.** One 135M model of one family and one 3B model of another; no scale study. At 135M the task is near the competence floor, the retention basket has little room to fall, and supervised targets must be filtered heavily before a verifier accepts them. RLV behaviour is known to differ across model families [5], and the Countdown model comes from exactly the family for which that was reported, so no cross-family generalisation is claimed.

**Arms are not matched on task performance.** The central comparison ends with the on-policy arm at 0.483 held-out satisfaction against 0.368 and 0.315 for the supervised arms. Prior work tests the retention account conditional on matched accuracy; ours is not matched and cannot be, because reaching parity would require sweeping the supervised arms over their budgets and selecting checkpoints. Joint dominance is therefore compatible with two explanations we cannot separate: the on-policy update is intrinsically more drift-efficient, or the supervised runs had not reached their efficiency frontier when stopped.

**Arms are not matched on supervision, steps or learning rate.** Shared: base model, prompt pool, split, verifiers. Not shared: training prompts ( $\approx 165$  versus 317), optimization steps ( $\approx 77$  versus 474), learning rate ( $5 \times 10^{-5}$  versus  $1 \times 10^{-5}$ ), and the form of supervision (dense token targets versus a scalar reward over sampled rollouts). The supervised arms need the higher rate to fit their targets at all, so equalising rates would have changed the data rather than improved the control.

**Replication.** Task skill and perplexity are measured over three training seeds; drift is a seed mean only for the on-policy arm; the commonsense probe is a single run per arm with no replication. No statistical test is reported for any drift or retention difference, because the per-seed quantities the test would need are not committed, and a test computed from recovered aggregates would test the digitisation rather than the experiment. The reported spread ( $\pm 0.023$  on skill,  $\leq 0.2$  points on perplexity) is an estimate from three samples, not a variance estimate.

**Drift is a length-normalised summary at base-policy contexts.** The sequence-level forward divergence is the sum of per-position divergences, whereas the reported quantity is a token average, so arms with different completion-length distributions are not compared on a reparameterisation of the same quantity; the artifacts do not contain per-arm evaluation-time lengths that would let us report the length-summed value alongside. Completions are sampled from  $\pi_0$ , so drift is measured on contexts the base model visits; behaviour on contexts that only a fine-tuned model reaches is not measured, and for a supervised arm that has moved substantially those contexts are a large part of its behaviour.

**Retention proxies.** Two perplexity corpora and one top-1 accuracy basket, none of which is a capability measurement. No capability-specific test with headroom was run, and the base model’s own corpus family had no small enough slice at this compute budget, so a mixed web/books/code sample stands in as a second opinion. The repository excluded GSM8K from the basket because the base score sits at a floor of about 1.4%—the right decision for an instrument, but it leaves the model’s weakest and most relevant skill unmeasured.

**Countdown evaluation is one-sided by construction.** The comparison log records a puzzle only when the trained model solves it, so the base model’s rate over the enumerated set is unmeasured and no paired statistic can be computed from that file. The 32/95 figure is a valid greedy solve rate over the puzzles the script enumerated, and the “base fails on all of them” statement is conditional on those same 32; the remaining 62 puzzles have no recorded verdicts, and the in-script scan reported further successes that were never written to the committed file, so 32 is a lower bound on what that checkpoint solved in that run.

**Known configuration defect.** The loader’s own warning in the committed log reports that the chat template’s end-of-turn string encodes to token id 151643 while the tokenizer’s `eos_token_id` is 151645, so the model is trained to emit a stop token on which generation need not halt. The effect is bounded by the logged clipping rate (2.8–5.3%) and is symmetric across base and trained evaluations, but it is a defect, and any follow-up should verify that template and generation configuration agree.

**Drift and reward definitions are measurement choices.** “GRPO” here denotes GRPO with a particular verifiable reward whose weights (2.0, 0.5, 0.2) were chosen after a farmable predecessor reward was abandoned; no reward ablation exists, because the earlier design was replaced rather than compared. Drift is reported in the forward direction  $D_{KL}(\pi_0 || \pi_\theta)$ ; the committed figure labels its axis with the reverse direction while its code computes the forward one, and every drift value in this paper uses the implemented direction, as Appendix A records.

**Provenance.** Per-run result files behind every aggregate are git-ignored, so values are recovered from committed figures by axis-calibrated digitisation ( $\pm 0.002$  on drift,  $\pm 0.01$  on skill) and from closed-form recomputation of the two committed Countdown logs. The checkpoints are not distributed, so independent verification requires retraining rather than re-measurement. One mismatch is flagged explicitly: skill and drift means come from seed-labelled runs, whereas the reported commonsense accuracy for the on-policy arm comes from a single run’s checkpoint-474.

**No mechanism ablation.** There is no reward sweep, no off-policyness sweep, no KL-coefficient sweep, no group-size sweep, no drift-matched supervised control and no large-k sampling to bound base-model coverage. Each is the experiment that would convert the inferences above into mechanistic findings; we list them as the highest-value follow-ups rather than as claims.

## 7 Conclusion

We asked how the post-training update rule interacts with distributional drift and with the retention of behaviour that was already present, using a reproducible small-scale artifact. The study contributes a controlled comparison in which three arms share one initialization, one prompt distribution, one held-out split and one verifier family; an exact full-vocabulary forward-KL measurement whose direction is fixed and justified; a demonstration that two retention probes of different sensitivity disagree about the same checkpoints; a negative result for the hypothesis that supervised drift is driven by the off-distributionness of targets; and a conservative analysis of a verifiable-reward reasoning gain in which the improvement is attributable to a search strategy the initialization already sampled.

The evidence supports these scoped statements. At the end of their training budgets, with task skill and perplexity averaged over three seeds, the on-policy arm reached the highest held-out instruction-following satisfaction ( $0.483 \pm 0.023$ ) while exhibiting the lowest measured forward drift ( $0.134 \pm 0.021$  nats per token against 0.331 and 0.334) and a perplexity change of +0.3% and +0.4% against +10.4–20.7% for the supervised arms, which were themselves indistinguishable in drift from each other. On the same checkpoints a commonsense multiple-choice battery placed every arm within one point of the base model, so whether degradation was detected was a property of the probe. In the Countdown experiment,

on-policy training raised the sampled solve rate from 9.1% to 21.9% over 500 steps and solved 32 of 95 enumerated held-out puzzles greedily, while 64% of the model’s own intermediate candidate equations remained arithmetically wrong.

The evidence does not support several statements that are easy to attach to these results, and we state them explicitly. It does not establish that on-policy RL forgets less than SFT in general, because the arms are not matched on task performance, steps, learning rate or supervision signal, only three seeds are available on some axes, and one model family per scale was tested. It does not establish a mechanism: the on-policy bias toward reweighting already-sampled behaviour is a hypothesis consistent with both experiments, not a result isolated by either. It does not demonstrate catastrophic forgetting in the strong sense, because no capability-specific test with headroom was run. And it does not show that reinforcement learning cannot create capabilities: the Countdown run shows a behaviour became stable and productive where the base model applied it badly, which is compatible with amplification and contains no test that would bound what the base model could already do.

Two small follow-ups would do most to sharpen these conclusions. A drift-matched supervised control—early-stopped or KL-regularised to the on-policy arm’s measured drift—would decide whether the retention gap belongs to the objective or to the distance moved. A large-k sampled evaluation of the Countdown base model on the same 95 puzzles would convert the amplification reading from an interpretation of logged chains into a measured bound on base-model coverage.

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## A Provenance of Every Reported Number

“Annotation” means the value is printed on the committed figure and read directly; “digitised” means it was recovered from the committed figure by gridline-calibrated axis mapping (linear-fit residuals below  $5 \times 10^{-4}$  in data units); “log” means it is computed from a committed raw log; “source” means it is a constant from the implementation, re-derived by re-running the listed script. No number in this paper comes from anywhere else.

Two checks were run on the digitisation: the linear fit that maps pixel coordinates to data coordinates, built from gridlines at known tick values, has a maximum residual of  $5 \times 10^{-4}$  in data units; and the recovered means reproduce the annotation strings printed on the same figures to within 0.002, which is why the annotation is preferred wherever both routes exist.

**Irrecoverable.** Three quantities used in the paper’s setup tables are reported with a caveat: (i) the number of matched prompts in the supervised comparison—the source prints  $\approx 165$  but writes the matched index list to a file that is not distributed; (ii) the learning rate of the DPO stage, inferred from the value the source states it settled on; and (iii) the wall-clock time of the Countdown run, estimated from the logged per-step time rather than taken from a recorded duration.

## B Claim Audit

### C Representative Model Outputs

The two chains below are the pair the repository highlights, on held-out puzzle index 25 (target 52, numbers {77, 33, 78, 86}); they are representative examples, not a statistical sample, and are reproduced because the aggregate numbers of §4 are unintelligible without them. Candidate lines of the form “expression = value (too high/low)” are annotated with their true values where the model’s own arithmetic is wrong.

**Base model (greedy).** The chain begins with a reasonable decomposition (“if we subtract 33 from 78, we get 45”), then enters a repetition loop in which the same sub-computation is restated with no progress: “*Let’s try: if we subtract 33 from 77, we get 44. Now we need to use 78 and 86 to get 8 ... We can use  $78 - 86 = -8$ , which is not helpful.*” repeats verbatim, and the model finally emits  $(78 - 86) + (77 - 33) = 52$ , whose left-hand side evaluates to 36. The structure of self-verification is present; it is not functional.

**Trained model (step 500, greedy).** The chain enumerates candidate expressions grouped by operator family and tags each with a judgement: “ $77 - 33 + 78 + 86 = 128$  (too high)” (true value 208), “ $77 * 33 / 78 + 86 = 33$  (too low)” (true value  $\approx 118.6$ ), then “*After trying various combinations, it seems*

**Table 5:** Provenance of the quantitative claims.

| Quantity                                                                                                     | Route      | Detail                                                                                                             |
|--------------------------------------------------------------------------------------------------------------|------------|--------------------------------------------------------------------------------------------------------------------|
| Task skill per arm (Table 3)                                                                                 | annotation | printed on the drift-skill figure, error bars $\pm 1$ sd, three seeds                                              |
| Task skill per-arm sd                                                                                        | digitised  | sds read from the same figure’s error bars                                                                         |
| Drift per arm                                                                                                | annotation | printed as “KL/tok” beneath each arm in the figure                                                                 |
| GRPO drift sd                                                                                                | digitised  | horizontal error bar, $\pm 0.021$ ; the supervised drift values carry no measurable bar                            |
| Perplexity deltas and sds                                                                                    | annotation | printed as “mean $\pm$ sd” per bar, three seeds                                                                    |
| Commonsense accuracy per arm                                                                                 | annotation | printed on that figure, single run per arm                                                                         |
| Base-model task skill                                                                                        | annotation | base point of the drift-skill figure                                                                               |
| Countdown solve rate, proximity, format, reward, KL, entropy, length, clipping, zero-variance group fraction | log        | logged trainer metrics, unmodified; derived values (e.g. solve rate = correctness reward/2.0) from the block table |
| Countdown held-out solves (32/95)                                                                            | log        | re-derived from the committed comparison log by slicing per puzzle header                                          |
| Countdown intermediate-arithmetic error rate                                                                 | log        | 18 of 28 parsed “expression = value” candidate lines mis-evaluate                                                  |
| Verifier coverage (20/25, split sizes)                                                                       | source     | re-derived by re-running the filter on <code>google/IFEval</code> (seed 42)                                        |
| Hyperparameters and budgets                                                                                  | source     | environment switches and per-run settings in the training-guide text                                               |

**Table 6:** Each row states a claim that appears in the paper together with its evidence class: **A** = directly measured by a committed artifact; **B** = inference supported by measurements within the tested configuration; **C** = plausible but not established by these experiments; **D** = unsupported, removed from the paper.

| Claim                                                                                                  | Class | Evidence                                                         |
|--------------------------------------------------------------------------------------------------------|-------|------------------------------------------------------------------|
| The three instruction-following arms share one initialization, prompt pool, split and verifier family. | A     | committed training scripts and configs                           |
| GRPO attains the highest held-out IFEval satisfaction at its endpoint ( $0.483 \pm 0.023$ ).           | A     | printed figure annotations, three seeds                          |
| GRPO exhibits the lowest measured forward per-token drift ( $0.134 \pm 0.021$ nats/token).             | A     | printed annotation plus digitised error bar                      |
| The two supervised arms are indistinguishable in drift ( $0.331$ vs. $0.334$ ).                        | A     | printed annotations, single measurement each                     |
| Supervised arms raise held-out perplexity by 10–21% while GRPO changes it by $\leq 0.4\%$ .            | A     | printed “mean $\pm$ sd” annotations on both corpora              |
| The multiple-choice battery separates no arm by more than one point from base.                         | A     | printed annotations, single run per arm                          |
| Countdown sampled solve rate rises $9.1\% \rightarrow 21.9\%$ ; held-out greedy solve rate is 32/95.   | A     | committed per-step log and comparison log                        |
| The trained model’s intermediate candidate arithmetic is largely incorrect (18/28).                    | A     | committed comparison log, recomputed                             |
| On-policy updates are constrained to reweight mass the policy already places.                          | C     | consistent with both experiments; not isolated                   |
| Drift is the causal driver of the measured retention differences.                                      | C     | would require a drift-matched supervised control                 |
| Retention cost is a property of the update rule as much as of the task.                                | B     | joint ordering under one shared task and evaluation              |
| The source of supervised targets does not drive this drift.                                            | B     | matched supervised arms differ by 0.003 nats/token               |
| Retention instruments measure different functionals and disagree.                                      | A     | both probes computed on identical checkpoints                    |
| Verifiable-reward training amplified rather than created the search behaviour.                         | B     | base chain already contains the structure; no large-k bound      |
| RL cannot create capabilities / RL forgets less than SFT (as universal claims).                        | D     | removed; the arms are unmatched on skill, steps, rate and signal |

we need to re-evaluate the approach”, followed by a second enumeration ending “ $77 - 33 - 78 + 86 = 52$  (this works)”, which is correct. Within this one chain there are 14 candidate lines and *nine* mis-evaluate their own expression—half of the 28 audited lines and half of the 18 errors behind the aggregate rate. Two expressions are asserted with two *different* values in the same chain:  $77 - 33 + 78 + 86$  is given as 128 early and as 208 later, and  $77 + 33 - 78 + 86$  as 112 early and 78 later. The reading is therefore not that the model cannot compute, but that enumeration and checking are present while the checks are applied inconsistently.

## D Reproduction Notes

The repository contains the data filters and verifiers, the supervised and on-policy training entry points, the drift measurement, the two retention probes, the figure scripts and both Countdown logs. The lockfile pins `trl` 1.8.0, `transformers` 4.57.6, `peft` 0.17.1 and `torch` 2.9.1; the 135M study fits a single 8 GB GPU in `bf16`, and the 3B run needs a 48 GB card. All randomness affecting the comparison is exposed through environment variables, with the split fixed at seed 42, the matched set at seed 0, and per-method seeds set separately. To reproduce Table 3 directly rather than from figures: run the three arms and the three on-policy seeds keeping the checkpoints; run the drift estimator once per checkpoint with the label naming convention the plot-

ting scripts expect; run both retention probes on the same checkpoints; regenerate the figures. The plotting scripts read `trainer_output/{kl,ppl,retention,grp,sft,ifeval_seeds}_runs.json`, so persisting those files is sufficient to reconstruct every aggregate without retraining, and Appendix A lists what remains unrecoverable even then. No training was performed while preparing this paper, and no committed number was regenerated; the evidence chain is source  $\rightarrow$  committed figures and logs  $\rightarrow$  calibrated extraction and closed-form recomputation  $\rightarrow$  tables.